

COLD FORMED HOLLOW SECTIONS

Other Tests			Dimensional Tolerances		
Flattening (min.)	Bend Test	Hydrostatic Test	On Diameter except for Non-Circular Tubes		On Wall Thickness
12	13	14	15		16
Apply for STD & XS of NB>2" Welded Portion: H= 2/3D The Other Side: H=1/3D	Apply fo NB ≤2" 12D at 90° When ordered for Close coiling: 8D at 180°	As per Table X2	1-1/2" & under	+ 1/64" - 1/32"	+ not limited -12.5%
--	--	1/2" to 1": P=700 psi 1-1/4" to 3": P= 1,000 psi 3-1/2" to 6": P=1,200 psi	2" & over	± 1 % of OD	
			1-1/2" & under	+ 1/64" - 1/32"	+ not limited -12.5%
			2" & over	± 1% of OD	
Welded Portion: H = 2/3D The Other Side: H = 1/3D	--	P=2St/D, Max. 2,500 psi S: Allowable Fiber Stress (16,000-18,000 psi) for Grade A. But in no case shall the stress produced exceed 80% of the specified Y.P	± 1% of OD		+ not limited -12.5%
Welded Portion: H = 2/3D The Other Side: H = 1/2D	--	--	Up to 1.90" 2" & over Under 2-1/2" 2-1/2" to 3-1/2" excluding 3-1/2" to 5-1/2" excluding 5-1/2" & over Square of Radii	± 0.5% ± 0.75% ± 0.02" ± 0.025" ± 0.03" ± 1% ± 2% max.	± 10%
Welded Portion: H = 2/3D The Other Side: H = 1/3D		P=2St/D S: Allowable Fiber Stress of 14,000 psi	As per Table 4 & 7		As per Table 6 & 8
--	6D at 90°	--	+ 0.001" - 0.005"		Nominal 0.04" - 0.072" Min. 0.036" - 0.068"
Max. 3t Max. 5t Max. 8t	--	--	1" & under Over 1" to 2" Over 2" to 2-1/2" Over 2-1/2" to 3" Over 3" to 3-1/2" Over 3-1/2" to 4" Over 4" to 4-1/2"	±0.004" ±0.005" ±0.007" ±0.008" ±0.010" ±0.012" ±0.015"	±8%
--	6D at 180°	--	Only 1.906" ±0.013" -0.017"		+ 20% - 10%
Apply for NB>2" Welded Portion: H=0.75D The Other Side: H=0.6D	Apply for NB≤2" Ungalvanized Tubes: 6D at 180°; Galvanized Tubes: 8D at 90°	P=700 lbf/in ² (50 kgf/cm ²)	Light Tubes:	Table 1	Light Tubes: -8%; + not limited
			Medium, Heavy Tubes:	Table 2	Medium, Heavy Tubes: -10%; + not limited
Max. 3t Max. 5t Max. 8t	--	--	1-1/4" & under Over 1-1/4" to 1-5/8"	+ 0.001" -0.005" +0.002" - 0.006"	±8%
H'= 3t or 1/2D H'= 6t or 3/4D H'= 8t or 7/8D H'= 6t or 3/4D (whichever is the smaller)	--	--	1" & under over 1" to 2" over 2" to 2-1/2" over 2-1/2" to 3" over 3" to 3-1/2" over 3-1/2" to 4" over 4" to 4-1/2" over 4-1/2" to 5" over 5" to 5-1/2" over 5-1/2" to 6" over 6" to 6-5/8"	±0.006" ±0.007" ±0.009" ±0.010" ±0.012" ±0.014" ±0.016" ±0.018" ±0.020" ±0.022" ±0.025"	±8%

*9: P = Test Pressure

*10 t¹ = Thickness of Flat Portion